

Answer the problems on **separate** paper. You do not need to rewrite the problem statements on your answer sheets. Do your own work. Show **all relevant steps** which lead to your solutions. Attach this question sheet to the front of your answer sheets.

1. (18 pts) For each of the following geometric series, determine:

- i. Does the series converge?
- ii. If the series is convergent, then find its sum.

a.
$$\sum_{n=1}^{\infty} 36 \frac{3^n}{4^{n+1}} = \frac{27}{4} + \frac{81}{16} + \frac{243}{64} + \cdots$$
 b.
$$\sum_{n=1}^{\infty} \frac{1}{12} \frac{4^{n-1}}{3^n} = \frac{1}{36} + \frac{1}{27} + \frac{4}{81} + \cdots$$

2. (30 pts) Determine whether the following series converge or diverge. Clearly identify which method you are employing, how you are applying your method and what your conclusions are.

a.
$$\sum_{n=1}^{\infty} \frac{1}{n^2 + 24n}$$
 b.
$$\sum_{n=1}^{\infty} \frac{n!}{n^3 6^n}$$
 c.
$$\sum_{n=1}^{\infty} 3(n+2) \left(\frac{2}{3}\right)^n$$

3. (12 pts) Find the first 3 non-zero terms of the MacLaurin series for $f(x) = \sqrt{9+x}$.

4. (24 pts) Find the first 4 non-zero terms of the MacLaurin series for:

a.
$$f(x) = \frac{\sin x}{x} - \cos x$$
 b.
$$f(x) = (1 - e^{-x})(1 + e^{-x})$$

5. (18 pts) Using a MacLaurin series expansion for $\cos x$,

- a1. approximate the value of $\cos 0.3$ by using the first 3 terms of the expansion
- a2. approximate the value of $\cos 0.8$ by using the first 3 terms of the expansion
- b1. find the maximum error which ensues by approximating the value of $\cos 0.3$ by the value obtained in a1.
- b2. find the maximum error which ensues by approximating the value of $\cos 0.8$ by the value obtained in a2.
- c1. from the maximum error in b1., determine the accuracy of the approximation in a1., i.e., determine how many decimal places are correct
- c2. from the maximum error in b2., determine the accuracy of the approximation in a2., i.e., determine how many decimal places are correct